BEST BRAIN EXAMINATIONS KONSORTIUM
SPECIAL PRIVATE MOCK EXAMINATIONS FOR BECE CANDIDATES – OCTOBER 2026
MARKING SCHEME - COMPUTING

QUESTION ONE

(a) (i) **FEATURE THAT MAKES DIRECT DATA ENTRY POSSIBLE**
Barcode

[2 marks]

(ii) **DEVICE USED TO CAPTURE STUDENT INFORMATION FROM THE ID CARD**
Barcode scanner / Barcode reader

[2 marks]

(iii) **REASON FOR RECOMMENDING A SOLID-STATE DRIVE (SSD)**
- Solid State Drives are faster.
- Programs and files open faster on Solid State Drives.
- Solid State Drives are more durable than HDDs.
- The lack of moving parts reduces the risk of mechanical failure.
- SSDs consume less power than HDDs.

[4 marks @ 2 marks each]

(b) (i) **IDENTIFYING THE VARIABLES IN THE PROGRAM**
- mysteryNumber
- correct
- guess

[2 marks @ 1 mark each]

(ii) **WHY THE IDENTIFIED VARIABLES WERE USED**
- mysteryNumber — used to hold the number which must be guessed.
- correct — It controls and ends the WHILE loop
- guess — Stores the students attempted number entered from the keyboard each time the loop runs.

[2 marks @ 1 mark each]

Academicprep.com
Learn • Practice • Excel

(iii) **LINES OF CODE DEMONSTRATING SEQUENCE, SELECTION, AND ITERATION**
-Sequence: Line 1 OR 2
-Selection: Line 5 to 7
-Iteration: Line 3 to 8

[3 marks @ 1 mark each]

(c) (i) **DEVICES DESCRIBED AS I TO VI**
I - Hearing aid (worn behind the ear)
II - AAC (Augmentative and Alternative Communication) device / picture-based communication app
III - Adaptive utensil handle / built-up grip aid (for weak grip)
IV - Prosthetic leg
V - Wheelchair
VI - Braille keyboard / Braille display

[3 marks @ ½ mark each]

(ii) **CLASSIFYING THE DEVICES INTO ASSISTIVE AND ADAPTIVE TECHNOLOGY**

ASSISTIVE TECHNOLOGY	ADAPTIVE TECHNOLOGY
I (Hearing aid) II (AAC/communication device) IV (Prosthetic leg) V (Wheelchair) VI (Braille keyboard)	III (Adaptive utensil handle)

[3 marks @ ½ mark each]

CamScanner

(iii) **BENEFIT OF EACH OF THE DEVICES IV AND V**

- IV (Prosthetic leg) — Helps individuals regain mobility, stand, walk, and participate in activities that might otherwise be difficult or impossible.
- V (Wheelchair) — Offers independence and flexibility for people with mobility limitations.

[3 marks @ 1½ mark each]

QUESTION TWO

(a) **HOW FRIMPONG WILL FIND THE EXACT LYRICS OF A GHANAIAN SONG**

(i) **Exact phrase search:**
Type the remembered line in quotation marks, e.g., "Ghana, the Black Star", in a search engine like Google to find webpages containing that exact phrase, increasing the likelihood of finding the song's lyrics.

(ii) **Wildcard operator:**
Use an asterisk (*) to represent unknown words, e.g., Ghana * Black Star, to search for variations of the phrase, allowing the search engine to fill in missing parts of the lyric.

[4 marks @ 2 marks each]

(b) **BENEFITS OF SOCIAL MEDIA IN EDUCATION**

- Enhancing communication between teachers and students, as well as between students themselves.
- Providing a platform for sharing information and resources among students and teachers.
- Encouraging collaboration and teamwork among students.
- Creating a more engaging and interactive learning environment through the use of multimedia.
- Supporting distance learning and remote education opportunities.

[2 marks @ 1 mark each]

(c) **BENEFITS OF WIKIS AND BLOGS IN COMMUNICATIONS**

- Multiple users can contribute and edit content, fostering teamwork.
- Easy dissemination of information to a wide audience.
- Enables one-to-many communication.
- Content variety.
- Feedback and engagement.

Academicprep.com
Learn • Practice • Excel

[4 marks @ 2 markS each]

(d) **THREATS THAT CAN PREVENT INFORMATION FROM REACHING ITS DESTINATION**

- Network congestion
- Malware/Viruses
- Physical damage to infrastructure
- Cyberattacks

[2 marks @ 1 mark each]

QUESTION THREE

(a) **HOW FORMULA ERROR OCCURS IN A SPREADSHEET**

A formula error in a spreadsheet occurs when ***a formula*** that has been entered ***cannot be calculated correctly*** or results in an invalid output due to incorrect syntax, referencing non-existent cells, division by zero, or incorrect data types
OR It is ***an error message*** shown when a ***spreadsheet formula*** is ***incorrectly written*** or has ***invalid data***. **OR** A formula error occurs when a ***spreadsheet formula cannot execute correctly*** due to ***invalid syntax***, ***incorrect cell references***, or ***mathematical issues,*** resulting in ***an error message***

[2 marks]

(b) **MEANING OF THE FORMULA ERRORS**

ERROR	WHAT IT MEANS
#DIV/0!	*The formula is attempting to divide a number by zero or an empty cell.*
#REF!	*The formula references a cell that is invalid, deleted, or unavailable.*
####	*The cell is too narrow to display the numerical result, or the result is a negative date/time.*

[3 marks @ 1 mark each]

CamScanner

(c) **CALCULATE 6 * 8 / 2 + (15 – 6) + 3^3**

6 * 8 / 2 + (15 – 6) + 3^3
= 6 * 8 / 2 + 9 + 27
= 6 * 4 + 9 + 27
= 24 + 9 + 27
= 60

[3 marks @ ½ mark each]

(d) **REASONS WHY VARIABLES SHOULD BE GIVEN MEANINGFUL NAMES**

- Makes code easier to read.
- Simplifies identifying issues.
- Helps in extending and updating code without confusion.
- Reduces dependence on comments.

[4 marks @ 2 marks each]

QUESTION FOUR

(a) **STEPS TO CREATE A HYPERLINK TO AN EXTERNAL WEBSITE IN A WORD DOCUMENT**

- Highlight the text or select the image to use as a hyperlink.
- Click on the Insert ribbon tab.
- Click on Link under the Links group.
- Enter the URL of the external website in the Address field.
- Click OK to confirm the hyperlink.

[5 marks @ 1 mark each]

(b) **DIFFERENCE BETWEEN PRESENTATION SOFTWARE AND DESKTOP PUBLISHING SOFTWARE**

- Presentation software is designed for creating slideshows for presentations, whereas desktop publishing software is designed for creating professional-quality layouts for print and digital publications.
- Presentation software focuses on slide transitions, animations, and multimedia integration, whereas desktop publishing software focuses on precise layout control, typography, and graphic design.

[2 marks @ 1 mark each]

Academicprep.com
Learn • Practice • Excel

(c) **PURPOSE OF CREATING MULTIPLE PARTITIONS ON A HARD DISK.**

It allows the separation of different types of data, such as system files and user files.

[2 marks]

(d) **ADVANTAGES OF PREDESIGNED TEMPLATES IN DESKTOP PUBLISHING SOFTWARE**

- Templates provide a quick starting point, reducing the time needed to create a document from scratch.
- Templates ensure a consistent, professional look and feel across multiple documents.
- Templates are easily customizable, allowing users to tailor content and layout to specific needs.
- Templates reduce errors by providing a structured framework for document creation.

[3 marks @ 1 mark each]

QUESTION FIVE

(a) **MEANING OF ENCRYPTION AND DECRYPTION AS APPLIED IN DATA PROTECTION.**

- Encryption is the process of converting plaintext data into cipher text using an algorithm and a key, making it unreadable to unauthorized users.
- Decryption is the process of converting cipher text back into plaintext using the appropriate key, allowing authorized users to access the original data.

[4 marks @ 2 marks each]

(b) **MEANING OF SOCIAL ENGINEERING**

Social engineering is a manipulation technique used by attackers to trick individuals into revealing sensitive information or granting access to secure systems

[2 marks]

CamScanner

(c) **MAIN DIFFERENCE BETWEEN AN IMPACT PRINTER AND A THERMAL PRINTER**

- An impact printer uses mechanical force to create text or images by physically striking an ink ribbon against paper.
- A thermal printer uses heat to transfer ink or create an image, either by heating special thermal paper or by melting wax/resin-based ink onto the paper, without mechanical impact

[3 marks @ 1½ mark each]

(d) **SAFETY MEASURES IN RISK REDUCTION AT WORKSTATIONS**

- Ergonomic Setup: Adjust chairs, desks, and monitors to maintain neutral postures (e.g., monitor at eye level, feet on the floor, wrists straight while typing).
- Proper Lighting: Ensure adequate, glare-free lighting to reduce eye strain. Use adjustable task lighting and positi monitors to avoid reflections.
- Cable Management: Organize and secure cables to prevent tripping hazards or electrical risks. Use cable ties or tr
- Regular Breaks: Encourage short breaks every 30-60 minutes to reduce fatigue and repetitive strain injuries. Prom stretching or standing.
- Anti-Slip Mats: Use non-slip mats in areas prone to spills or where standing work is common to prevent slips and falls.
- Fire Safety: Keep workstations free of clutter and ensure easy access to fire exits. Install smoke detectors and fire extinguishers nearby.
- Ventilation: Maintain good air quality with proper ventilation to prevent respiratory issues or discomfort from dust fumes, or stale air.

[3 marks @ 1½ mark each]

PAPER ONE- OBJECTIVE TEST [40 MARKS]

Acadmicprep.com
Learn • Practice • Excel

1. B	11. A	21. A	31. D
2. B	12. B	22. B	32. B
3. C	13. B	23. B	33. A
4. B	14. B	24. B	34. C
5. D	15. A	25. D	35. C
6. B	16. B	26. B	36. B
7. B	17. C	27. C	37. D
8. B	18. B	28. C	38. B
9. B	19. C	29. B	39. B
10. C	20. B	30. C	40. C

[40 marks]

CamScanner